

Q2_uniform_quadratic_bspline

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1 ISyE 6525 HW1 - Question 2

For the uniform knot vector

$$\{\tau_0, \tau_1, \tau_2, \tau_3, \tau_4, \tau_5\} = \{0, 1, 2, 3, 4, 5\},$$

the Cox-de Boor recursion is

$$B_{i,j}(x) = \frac{x - \tau_i}{\tau_{i+j} - \tau_i} B_{i,j-1}(x) + \frac{\tau_{i+j+1} - x}{\tau_{i+j+1} - \tau_{i+1}} B_{i+1,j-1}(x),$$

with

$$B_{i,0}(x) = \begin{cases} 1, & i \leq x < i+1, \\ 0, & \text{otherwise.} \end{cases}$$

1.1 Step 1: degree-one basis functions

Since the knots are one unit apart,

$$B_{i,1}(x) = (x - i)B_{i,0}(x) + (i + 2 - x)B_{i+1,0}(x).$$

The degree-one functions needed below are

$$\begin{aligned} B_{0,1}(x) &= \begin{cases} x, & 0 \leq x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{otherwise,} \end{cases} & B_{1,1}(x) &= \begin{cases} x - 1, & 1 \leq x < 2, \\ 3 - x, & 2 \leq x < 3, \\ 0, & \text{otherwise,} \end{cases} \\ B_{2,1}(x) &= \begin{cases} x - 2, & 2 \leq x < 3, \\ 4 - x, & 3 \leq x < 4, \\ 0, & \text{otherwise,} \end{cases} & B_{3,1}(x) &= \begin{cases} x - 3, & 3 \leq x < 4, \\ 5 - x, & 4 \leq x < 5, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

1.2 Step 2: derive $B_{0,2}(x)$

Using the recursion,

$$B_{0,2}(x) = \frac{x}{2}B_{0,1}(x) + \frac{3-x}{2}B_{1,1}(x).$$

Evaluate it on each interval:

$$B_{0,2}(x) = \begin{cases} \frac{x}{2}(x) = \frac{x^2}{2}, & 0 \leq x < 1, \\ \frac{x}{2}(2-x) + \frac{3-x}{2}(x-1) = -x^2 + 3x - \frac{3}{2}, & 1 \leq x < 2, \\ \frac{3-x}{2}(3-x) = \frac{(3-x)^2}{2}, & 2 \leq x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

1.3 Step 3: derive $B_{1,2}(x)$ and $B_{2,2}(x)$

For $B_{1,2}$,

$$B_{1,2}(x) = \frac{x-1}{2}B_{1,1}(x) + \frac{4-x}{2}B_{2,1}(x),$$

so

$$B_{1,2}(x) = \begin{cases} \frac{(x-1)^2}{2}, & 1 \leq x < 2, \\ -(x-1)^2 + 3(x-1) - \frac{3}{2}, & 2 \leq x < 3, \\ \frac{(4-x)^2}{2}, & 3 \leq x < 4, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly,

$$B_{2,2}(x) = \frac{x-2}{2}B_{2,1}(x) + \frac{5-x}{2}B_{3,1}(x),$$

which gives

$$B_{2,2}(x) = \begin{cases} \frac{(x-2)^2}{2}, & 2 \leq x < 3, \\ -(x-2)^2 + 3(x-2) - \frac{3}{2}, & 3 \leq x < 4, \\ \frac{(5-x)^2}{2}, & 4 \leq x < 5, \\ 0, & \text{otherwise.} \end{cases}$$

Because the knots are uniform, these are translated copies:

$$B_{1,2}(x) = B_{0,2}(x-1), \quad B_{2,2}(x) = B_{0,2}(x-2).$$

```
[1]: import numpy as np
import matplotlib.pyplot as plt

def B02(x):
    x = np.asarray(x, dtype=float)
    return np.select(
        [(0 <= x) & (x < 1),
         (1 <= x) & (x < 2),
         (2 <= x) & (x < 3)],
        [x**2 / 2,
         -x**2 + 3*x - 3/2,
         (3-x)**2 / 2],
        default=0.0,
    )

def B12(x):
    return B02(np.asarray(x) - 1)

def B22(x):
    return B02(np.asarray(x) - 2)

x = np.linspace(0, 5, 1001)

plt.figure(figsize=(8, 4.5))
plt.plot(x, B02(x), label=r'$B_{0,2}(x)$', linewidth=2)
plt.plot(x, B12(x), label=r'$B_{1,2}(x)$', linewidth=2)
plt.plot(x, B22(x), label=r'$B_{2,2}(x)$', linewidth=2)
plt.xticks(range(6))
plt.xlim(0, 5)
plt.ylim(0, 0.82)
plt.xlabel('$x$')
plt.ylabel('Basis value')
plt.title('Uniform quadratic B-spline basis functions')
plt.grid(alpha=0.25)
plt.legend()
plt.tight_layout()
plt.show()
```

